

TrueSelect©
Domain Analysis
Final Report

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Introduction

In the U.S., tech companies spend over \$4,000 per hire¹². These costs arise from various inefficiencies, including evaluating dozens or even hundreds of applications for each job opening, covering HR expenses, and allocating resources for training new hires.

Additionally, applicants often lack a way to define their chances of success in advance, encouraging them to apply to multiple positions simultaneously in the hope of securing at least one offer. The hiring process is further complicated by human bias and superficial judgments, which can result in suboptimal hires—an especially critical issue in the highly competitive tech industry. Companies in this market require rapid development to stay on top of the field, for which a strong and expanding workforce is necessary.

In the rapidly evolving landscape of technology job recruitment, the integration of artificial intelligence (AI) has introduced both opportunities and challenges. Recruitment processes, historically plagued by inefficiencies and implicit biases, are now being reimaged through AI-driven solutions. These innovations promise not only to streamline the hiring process but also to enhance fairness by focusing on purely objective criteria. However, the adoption of such systems raises critical questions regarding transparency, ethical standards, and stakeholder satisfaction.

This report provides a comprehensive domain analysis of AI recruitment technologies, focusing on their implications for the tech industry. It explores how these tools align with the goals of diverse stakeholders, including large tech companies, job seekers, and regulatory bodies. By examining the game-theoretic dynamics, biases, and legislative considerations inherent in these systems, this report seeks to identify strategies that balance efficiency, fairness, and trust in building recruitment systems that can meet the complex demands of a dynamic labor market.

Problem Perspective

Before exploring the problem domain and outlining the proposed solution, it is vital to highlight the definition of the problem first. In principle, this problem should be defined as a **social conflict problem** - meaning that, in principle, it is impossible to eliminate. This happens as the conflict arises between maximally optimizing the hiring procedure while allowing equal opportunities to the applicants.

Matchmaking inherently favors specific traits, as seen in sports, where players with strong mental resilience often outperform technically superior opponents. Similarly, in hiring, there is **no definitive ground truth** for identifying the "best" candidate. Our approach aims not to achieve a perfect solution but to minimize this social conflict by providing tools that empower both applicants and hiring departments to make better, more informed decisions.

This inherent conflict will resurface later in the report, illustrated by the well-known example of Amazon's failed recruitment tool³, discussed in the section on Current AI Applications. For now, we turn to a key question: who stands to benefit from addressing this problem, and how?

Stakeholders

In this section, we shift the focus from the problem itself to the stakeholders affected by it in order to clearly define the amplitude and scale our solution applies to. This section is ordered hierarchically, meaning we first start with the most important (primary) stakeholder and then highlight the other parties included, sorted by the strength of impact.

¹<https://engagedly.com/blog/average-cost-per-hire-employee/>

²<https://brocoders.com/blog/how-much-do-tech-companies-spend-on-recruiting/>

³<https://www.reuters.com/article/world/insight-amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK0AG>

Primary Stakeholder

The primary stakeholders considered by the team are **large tech companies**, as they will bear the cost of implementing this system and have the most to gain from improving their hiring processes. These companies strive to increase productivity by hiring the best-fit candidates while minimizing the costs, as mentioned in the Introduction. Their *desired output* includes a detailed, ranked list of applicants, supported by explanations such as their strengths and weaknesses. By automating the ranking process, the system would significantly reduce the workload for HR departments, allowing them to reallocate resources to higher-value activities, such as employee retention strategies, ultimately enhancing workplace satisfaction.

The *stakeholder's intention* is that the choice of the candidate should be based on an unbiased personality evaluation, combined with skills in relation to the tech position. This approach shifts the bias of human selection into the bias of the AI system (described further in the Granularity Analysis section), which can be continuously monitored and mitigated—unlike human biases, often undetectable and unchangeable. This is why we believe this stakeholder to be positively impacted by this system.

Although large tech companies are the primary focus, as they are the ones funding and deploying the system, the outputs they require are also highly relevant to another group: job applicants.

Secondary Stakeholders

(a) Tech Job Applicants

The detailed ranking description mentioned in the previous section could be published to become the **desired output** for this group as well. This approach aligns with the *stakeholder's intention* - relieving stress, ensuring the final evaluation process is fair, and understanding areas of improvement. These rankings would provide applicants with clear feedback on which skills to develop to perform better in similar applications, ultimately improving the overall quality of candidates.

By discouraging underprepared applicants or motivating others to upskill before reapplying, this system addresses the issue of excessive applications per position (outlined in the Introduction). At the same time, incorporating this stakeholder into the solution takes no effort as those rankings are already developed for the *primary stakeholder*, meaning that implementing this system would improve hiring for companies and enhance transparency and fairness for applicants.

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(b) People Looking to Switch Industries

With the rise of IT and AI combined with relatively high payslips, an increasing number of young professionals decide to switch their current professions, such as learning Python to become a software engineer. This leads to an unnecessary saturation of the market which hurts both the employers and the employees. Additionally, career switchers face significant risks, investing time and resources into learning skills unrelated to their current expertise, only to find they may still struggle to secure their desired roles due to competition with more specialized candidates.

To address this issue, the AI interviewer could be published (e.g. on a pay-per-request basis) to provide support for "*what if*" analysis⁴. By allowing individuals to create fictional CVs—including future certifications and skills they intend to acquire—the AI interviewer could evaluate their potential for a specific role.

⁴<https://support.microsoft.com/en-us/office/introduction-to-what-if-analysis-22bffa5f-e891-4acc-bf7a-e4645c446fb4>

The *stakeholder's intention* here is to gain insight into whether investing effort in upskilling would likely result in landing their desired job. This would perfectly align with the **desired output**, which would consist of ranking the potential CV and highlighting its strengths and weaknesses, enabling informed decisions on whether to proceed with the career switch.

(c) Legislators / regulatory bodies

Due to the AI act being in almost full force, this stakeholder is vital to consider before deploying and using our model. The importance of this forces the issue to be considered explicitly, which will be done in the Legislation Issues and Compliance section. What is relevant for now is to keep in mind that all the outputs and decisions made by the model have to align with this stakeholders **desired output**: explainability and interoperability.

The *stakeholder's intention* is that the model should treat applicants fairly and avoid biases to ensure the use of the system is truly beneficial for society. Importantly, those regulations are highly beneficial in a manner that they will indirectly force a high quality of the model and AI practices applied in the project.

This section provided a thorough analysis of the stakeholders, combined with their intentions and desired outputs. Now, we switch the perspective on those stakeholders by clearly defining their goals and strategy for achieving them to put them in the context of the *social conflict problem*, as outlined in the previous section.

Game-Theoretic Approach for Stakeholder Analysis (Talajić et al., 2024)

The interactions between stakeholders in the recruitment system can be analyzed through game-theoretic lens, providing a framework to understand their goals, strategies, and trade-offs. This approach offers insights into how fairness, efficiency, and trust can be balanced in the system.

Large tech companies, as the primary stakeholders, aim to maximize productivity by identifying and hiring candidates who best fit their organizational needs. They also seek to streamline the hiring process to reduce costs and inefficiencies while maintaining fairness and compliance with legal standards. Job applicants, as secondary stakeholders, prioritize fair and transparent evaluations, along with actionable feedback to enhance their skills and employability. Regulators, representing another group of secondary stakeholders, focus on ensuring that AI recruitment tools adhere to ethical and legal standards, promoting equity and preventing systemic biases.

Companies must decide on the extent of AI integration, ranging from full automation to hybrid systems, and determine the balance between technical skills, personality fit, and diversity in their evaluation criteria. Applicants, on the other hand, face choices such as applying broadly or selectively based on the perceived fairness of the system, using feedback to improve future applications, or contesting unfavorable outcomes.

The payoffs for companies involve a combination of hiring efficiency, candidate fit, and reputational impact, offset by the costs associated with compliance and potential biases. For applicants, the payoffs include their probability of selection and the transparency benefits, balanced against the stress and costs associated with the application process.

Optimal outcomes in this context involve alignment among stakeholders on fairness, efficiency, and transparency, resulting in mutual satisfaction and improved hiring outcomes. Conversely, failure to address biases and inefficiencies could lead to distrust, legal penalties, and reduced effectiveness of the recruitment system.

By adopting this game-theoretic perspective, the complexities of stakeholder interactions can be better understood, guiding the development of recruitment systems that align with both organizational goals and societal expectations.

Undesirable predictions/outcomes (prejudice)

Understanding Prejudices in Recruitment Processes

Bias in recruitment is an issue in the tech industry, often favoring candidates based on superficial or non-performance-related factors, such as appearance, gender, or affiliations. These prejudices result in single-minded teams that lack diverse perspectives, limiting innovation and excluding highly qualified individuals that don't fit the traditional mold of a person in tech. Addressing this requires rethinking recruitment systems. Traditional methods, which rely heavily on human judgment, are open to implicit prejudices that the recruiter may have. They often overlook candidates with unconventional backgrounds or those from underrepresented groups who might excel in technical roles if evaluated fairly. Recognizing these systemic flaws is the first step in designing AI systems to eliminate such biases.

AI as a Tool for Breaking Prejudices

AI models can help recruitment by focusing on actually good recruitments rather than what has traditionally been seen as a 'Good hire'. By anonymizing candidate data during the screening process, AI can prevent recruiters from being influenced by demographic or personal information. Additionally, AI can identify patterns of bias in historical hiring data, allowing companies to rethink their recruitment strategies and possibly even re train (educate) people with the new data. Through regular auditing and actively avoiding bias, these models can fairly look through bigger talent pools and ensure every candidate has an equal chance of being selected based on their potential to excel in the role.

Current AI applications

- (a) AI in talent acquisition: a review of AI-applications used in recruitment and selection ⁵
- (b) The Power of Artificial Intelligence in Recruitment: An Analytical Review of Current AI-Based Recruitment Strategies ⁶
- (c) Ethics of AI-Enabled Recruiting and Selection: A Review and Research Agenda ⁷
- (d) AI-enabled recruiting: What is it and how should a manager use it? ⁸
- (e) Insight - Amazon scraps secret AI recruiting tool that showed bias against women ⁹

Desk Research Summary

Existing research indicates that Artificial intelligence (AI) is increasingly being adopted in recruitment and selection (R&S), offering potential for efficiency and innovation but also raising significant challenges. Current applications of AI in R&S focus on areas like chatbots for initial communication, screening software for resumes, and automation tools for routine tasks that can be encountered in the recruitment process. These technologies can streamline the R&S pipeline, thus reducing costs and expanding the candidate pool, which consequently can result in increased productive output. However, overall investment in AI for R&S is still tentative and adoption remains concentrated, in large, to innovative firms and early-stage start ups.

⁵<https://www.emerald.com/insight/content/doi/10.1108/shr-04-2019-0024/full/html>

⁶<https://openaccessjournals.com/JBReview/article/view/2089>

⁷<https://link.springer.com/article/10.1007/S10551-022-05049-6>

⁸<https://www.sciencedirect.com/science/article/pii/S0007681319301612>

⁹<https://www.reuters.com/article/world/insight-amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK0AG/>

Despite its promise, AI in recruitment poses critical issues. For example, Amazon’s failed AI recruiting tool revealed how machine learning can perpetuate biases when trained on historical, skewed data. In 2014, Amazon developed an AI-powered recruitment tool aimed at automating the hiring process. However, by 2015, the company discovered that the tool was biased against female applicants. This bias stemmed from the model’s training data, which comprised résumés submitted to Amazon over a decade—predominantly from male candidates, reflecting the tech industry’s male dominance. The AI system learned to favor male candidates and downgraded résumés containing terms associated with women, such as references to all women’s colleges or activities with the word “women’s.” Despite attempts to adjust the algorithm to be gender-neutral, concerns persisted that it might find other ways to discriminate against female candidates. This incident highlighted the risks of algorithmic bias and discrimination, demonstrating how AI systems can perpetuate existing societal biases present in their training data. Consequently, Amazon discontinued the use of this recruitment tool.

While AI tools like HireVue and LinkedIn offer innovative solutions to match candidates to roles, **experts emphasize that AI should augment rather than replace human judgment.** The technology is still evolving, requiring robust data practices, regulatory oversight, and ethical safeguards to achieve fairness and trust in recruitment processes.

Existing AI Solutions In Recruitment

The recruitment pipeline can be roughly cut into three separate stages: **sourcing, screening, and the hiring process.** Various automation techniques have been applied largely to all three stages.

- (a) **Sourcing:** AI is all about saving time by automating the more predictable and repetitive tasks of any practice. Most of these tools have remarkable sourcing capabilities. They can scour all kinds of corners of the internet for candidates, or provide few really specific profiles. Many also do smart matches and recommendations based on a few inputs.
- (b) **Screening:** As you work that candidate pipeline, the same tools tend to have different ways of filtering people. Our software selection includes tools that incorporate AI into assessments, interview scheduling, and screening, whether it’s through chatbots or smart shortlisting.
- (c) **Hiring Processes:** From AI-generated contracts and offer letters to AI assistants that help with onboarding, many of these tools put a cherry on top by also providing features for the very last step of the recruiting process.

It is important to note that **the quality of the hired employees** (and thus the quality of the production output) **is dependent on the success in all three separate stages.** This means that even if the screening process implements AI-based filtering, but the sourcing of the candidates lacks proper scope or the hiring process itself is struggling to convert accepted candidates into employees then the system all together fails to accomplish the goal of hiring thriving workers. It means that all three stages should be looked at as a whole in order to build create an automated recruitment and selection pipeline. This can also be seen in AI-based R&S tools that are offered in the industry - rather than offering a pure model to filter and select candidates, over-arching software solutions are offered instead that automate/enhance one or more of the three processes in the R&S pipeline.

Current industry standard AI R&S tools and companies

This collection of companies is meant to give an overview of the competition and is not an exhaustive list.

- (a) 1. Humanly: Designed for mid-market companies, Humanly automates candidate screening, scheduling, and engagement through an AI-powered chatbot. It integrates with various Applicant Tracking Systems (ATS) and uses data from performance management systems to en-

hance screening accuracy. However, it lacks features like candidate rescheduling and interview reminders, and its pricing is not publicly disclosed.

- (b) 2. **Paradox:** Paradox’s AI assistant, Olivia, interacts with candidates via chat, automating tasks such as screening, interview scheduling, and answering candidate queries. It’s particularly effective for high-volume hiring and integrates seamlessly with existing systems. A potential drawback is that its advanced features may be more than necessary for smaller organizations.
- (c) 3. **HireVue:** HireVue offers AI-driven video interviewing and assessment tools that evaluate candidates’ competencies and fit. It’s suitable for large enterprises aiming to streamline their screening process. Concerns have been raised about potential biases in AI assessments, and the platform’s complexity might require significant training for optimal use.
- (d) 4. **Pymetrics:** Pymetrics utilizes neuroscience-based games and AI to assess candidates’ cognitive and emotional attributes, aiding in unbiased screening. It’s beneficial for organizations focusing on diversity and inclusion. However, some users may find the gamified assessments unconventional, and the effectiveness of such evaluations can vary across different roles.

Additional list of AI R&S companies

- (a) **Workable:** AI recruiting tool with video interviews HR capabilities
- (b) **Humanly:** Best for Mid-Market Companies
- (c) **Paradox:** AI focus on high-volume hiring
- (d) **Manatal:** SMB-friendly AI-powered ATS
- (e) **Wellfound:** AI sourcing with human oversight for scaling startups
- (f) **AmazingHiring:** Best for Sourcing Tech Candidates scaling startups
- (g) **hireEZ:** Best for Sourcing Talent Insights
- (h) **HireVue:** AI recruiting for large enterprises
- (i) **Fetcher:** Highly specialized AI Sourcing Outreach Tool
- (j) **Eightfold:** AI Recruiting + Coaching Skills Development
- (k) **Findem:** AI sourcing with human oversight for scaling startups
- (l) **Textio:** Best for Recruitment Marketing Optimization

Ground truth analysis

- **Presumed ground truth:** Person X is the best candidate for a job Y
- **Alternative views:** For some jobs, even a person who does not suit in quite well can learn fast and become a better technician than a candidate in the long run. Other than that, a good worker who will stay loyal to the company for multiple years to come is better than a perfect candidate who will leave after a year. Moreover, a good worker who is very dedicated to the job in the long term is better than a perfect candidate who will not care about the job too much and only do what is required of them (if even). Lastly, the model would have to be very flexible in the choice of variables i.e. for some technical work it should favor candidates that work well in large groups while others prefer pure talent, which can lead to ambiguous decisions in edge cases.

- **Hegemonic perspective:** not existent. No ground truth is currently possible. (as mentioned in the Problem Perspective section)

Granularity Analysis

- High-level: Choosing a candidate from a list. That would simplify the hiring process while not requiring any extra processing from a human employer. Problems - sometimes two candidates (or more, or none) are suitable for the job, and this discards that fact + useless for the applicants
- Mid-level: Assigning each candidate a score. This would allow for a clear comparison between the candidates and would facilitate the best allocation of resources, allowing the employer to choose how many of the applicants should get the job. Problems - does not allow for direct comparison between candidates, as a standard sensitivity analysis will not cover all differences between the scores obtained by the candidates + useless for the candidates.
- Low-level: assigning each candidate a score combined with a detailed explanation/score distribution over domain i.e. expertise, soft-skills, fitness, etc. This would allow for a thorough and hopefully unbiased analysis of the candidates, allowing for the choice based on a fair comparison determined by the true requirements of the employer. Would also be useful for the applicants to see why they haven't been selected. Problems: would require a lot of computational resources for training and would still require a lot of work from the recruiters as all applicants would probably have their strengths and weaknesses
- Very-low-level: combine the score and the detailed description with a comparison between other candidates based on a position. It would be very useful in relieving the necessity of human comparison, fully automating the process by saying e.g. "Candidate A is the best overall, but Candidate B is expected to improve the current team's efficiency and chemistry more than Candidate A". That would allow the recruiters to choose candidates based on not only their profiles but also on multiple aspects related to the particular place they are supposed to be in a company with a relation to each other. Problems - almost impossible not to make this system biased.

Useful Variables

This section outlines suggested variables for consideration in the development of AI models for recruitment. These variables are indicative and represent potential inputs that could influence candidate evaluation; however, the specific features implemented in a model should be carefully tailored to align with the unique requirements of the company, position, and field of recruitment. Generalizing variables across industries or roles may fail to capture critical nuances essential to fair and effective company-specific hiring decisions. The variables presented are organized into three broad categories: educational, work experience, and personal attributes. Educational variables, such as the highest degree obtained, GPA, and field of study, provide insights into a candidate's academic background and qualifications. Work experience variables, including previous job titles, technical skills, and turnover rate, help evaluate a candidate's professional expertise and career trajectory. Lastly, personal attributes, such as interpersonal skills, volunteering activities, and safety screening, assess aspects related to cultural fit and reliability. All of the suggested variables should be moderately accessible either through CV-screening, public information on recruitment platforms (e.g., LinkedIn) or pre-screening questionnaire forms.

Variables

The variables are summarized in Tables 1, 2 and 3.

Variable	Explanation	Power
Highest Degree Obtained	-	Medium
Grade Point Average	-	Low-Medium
University Ranking	-	Low-Medium
Field of Study	Relevance to the job applied	High
Extracurricular activities	-	Medium
Certificates/Qualifications	Official certifications besides degree	High
Awards	-	Medium

Table 1: Educational Variable Description

Variable	Explanation	Power
Previous Job Titles	Relevance of previous positions	Very High
Work Experience	-	Very High
Other technical Skills	-	Medium
Turnover rate	Average time spent per job	High
Job-specific keywords	Amount of Job Specific Keywords in CV	High

Table 2: Work Experience Variable Description

Variable	Explanation	Power
Interpersonal Skills	Presentation, leadership, communication and teamwork skills	Low
Safety Screening	Background check, high if problematic	High
Personal interests	Personal Interests to align with company culture	Low
Volunteering activities	Pro-Bono Work, related or unrelated to field	Very Low

Table 3: Personal Variable Description

Bias in AI Recruitment Systems

Bias in recruitment is an unavoidable challenge, whether hiring decisions are made by humans or AI systems. Human recruiters bring implicit biases shaped by personal experiences and social norms, which can lead to discrimination—both intentional and unintentional. While AI can help mitigate some of these biases by applying consistent criteria, it is not inherently free of bias. Instead, AI models learn from historical hiring data, which may reflect past discrimination, reinforcing existing inequalities rather than eliminating them.

Bias in AI-driven recruitment appears at three key levels. *Data-level bias* arises when training datasets reflect systemic inequalities, leading models to replicate these patterns. *Model-level bias* can emerge from the way features are weighted, where certain traits (such as university prestige) may be overemphasized, unintentionally disadvantaging otherwise qualified candidates. *Outcome-level bias* occurs when AI-generated rankings favor specific demographics over time, creating feedback loops that reinforce existing disparities. If AI consistently prioritizes a narrow subset of candidates, future training data will reflect these hiring trends, amplifying bias rather than correcting it.

To address these risks, AI-driven hiring systems must incorporate structured bias mitigation strategies. Before model training, datasets should be diversified and pre-processed to minimize

inherited biases. During training, fairness-aware algorithms should ensure that no demographic group is unfairly penalized. After deployment, regular fairness audits, demographic impact assessments, and human oversight must be integrated to detect and correct biases in real-world applications. Bias cannot be fully removed, but it can be systematically reduced through ongoing monitoring and recalibration.

Granularity as a Bias Mitigation Strategy

One way to improve fairness in AI recruitment is through greater granularity in candidate evaluation. Traditional AI-based hiring models often produce a single ranked list, condensing complex evaluations into a single score. This approach risks masking biases embedded in the model's decision-making process. A more granular system, where candidates are assessed across multiple dimensions—such as technical skills, soft skills, cultural fit, and experience—provides greater transparency and allows for targeted bias corrections. If disparities emerge in a particular category, adjustments can be made to improve fairness without distorting overall rankings.

However, increased granularity comes with trade-offs. More detailed evaluations require greater computational resources and can introduce overfitting risks, where AI models learn to replicate historical hiring patterns rather than adapt to changing workforce needs. Additionally, while breaking down scores into subcategories improves interpretability, excessive complexity may make hiring decisions harder to understand for both applicants and employers. To balance these concerns, AI-driven hiring systems should focus on meaningful differentiation rather than excessive subcategorization, ensuring that granular evaluations enhance fairness without overwhelming decision-makers.

Addressing Feedback Loops and Long-Term Bias Risks

A major risk in AI-driven recruitment is the reinforcement of hiring biases over time. If an AI system consistently favors certain profiles—such as graduates from top-ranked universities or candidates with specific past job titles—future hiring decisions will further skew towards these groups, limiting diversity and innovation. This feedback loop not only reduces fairness but also makes the AI system less effective by narrowing its candidate selection over time.

To counteract this, AI recruitment models must be designed with dynamic bias correction mechanisms. Training data should be regularly updated to reflect evolving industry trends, ensuring that hiring decisions do not become self-reinforcing. Fairness-aware ranking algorithms should be used to prevent overconcentration of certain candidate profiles. Additionally, applicant responses to AI-generated evaluations should be monitored to detect potential system gaming, where candidates adjust their resumes to fit AI preferences rather than genuinely improving their skills.

By integrating dynamic fairness corrections, regular audits, and transparent evaluation criteria, AI-based hiring systems can become more equitable while maintaining their effectiveness in selecting top candidates.

Use-case analysis (scenario analysis)

In developing AI-driven recruitment tools like TrueSelect[®], conducting scenario analysis is a useful tool. This method involves envisioning and evaluating various potential situations to understand how the system performs under different conditions. By simulating specific use cases, we can identify potential challenges, assess system responses, and ensure that the tool operates effectively across diverse scenarios. Scenario analysis enhances predictive modeling by allowing organizations to visualize how changes in variables can influence outcomes, thereby improving decision-making and preparedness.¹⁰

¹⁰<https://www.nobledesktop.com/learn/data-analytics/the-benefits-of-scenario-analysis-in-predictive-modeling>

Automated Resume Screening

The first use case focuses on automating resume screening to enhance efficiency in the initial stages of recruitment. In traditional hiring processes, HR managers must manually review hundreds of resumes, consuming valuable time and increasing the likelihood of overlooking qualified candidates. With TrueSelect©, the AI-driven system streamlines this process by automatically analyzing resumes and ranking candidates based on predefined criteria. The primary actors in this use case include the HR Manager at a tech company, who is responsible for utilizing the system, and job applicants who submit their resumes for evaluation.

Before the screening begins, job applicants submit resumes through an online portal or integrated third-party platforms. The AI model is pre-trained on industry-specific hiring data, and the HR Manager configures screening parameters such as job-specific skills, educational background, and certifications. Once the process is initiated, TrueSelect© parses resumes, extracts key attributes, and assigns scores based on a weighted ranking system. The system then generates a list of top-ranked candidates along with a summary of their strengths. The HR Manager reviews this shortlist and refines it for interview invitations. If a resume lacks critical information, the AI flags it for manual review. Additionally, periodic bias audits ensure fairness and compliance with diversity policies. By automating this step, HR managers can focus their time on engaging with high-potential candidates, improving both efficiency and hiring outcomes.

AI-Driven Interview & Assessment

After resume screening, TrueSelect© conducts AI-driven preliminary interviews to evaluate candidates efficiently. This use case reduces the need for human intervention in early-stage interviews while ensuring a fair and consistent evaluation process. The primary actors include job applicants who participate in AI interviews and the HR Manager and technical team, who review AI-generated reports to make informed decisions.

Shortlisted candidates receive an invitation to the AI-driven interview. TrueSelect© is configured with assessment questions for both behavioral and technical evaluation. The interview platform supports video, audio, or text-based interactions, depending on the candidate's preference. The system administers structured questions, records responses, and analyzes them based on speech patterns, facial expressions (if video-based), problem-solving approach, and technical correctness. Once the interview is completed, TrueSelect© generates a comprehensive assessment report highlighting strengths, weaknesses, and overall performance. The HR Manager and technical team review the results to determine the next steps. Candidates who opt out of AI interviews are provided with alternative assessment options, ensuring flexibility in the evaluation process. System logs are maintained to track any discrepancies in responses, allowing HR teams to conduct further reviews if necessary.

Final Candidate Ranking & Feedback Generation

The final stage of the recruitment process involves ranking shortlisted candidates and providing transparent feedback to both HR and applicants. This ensures fairness and allows unsuccessful candidates to improve for future applications. The HR Manager acts as the primary decision-maker, while job applicants benefit from receiving structured feedback on their performance.

Once all assessments are complete, TrueSelect© consolidates the data to create a final ranking of candidates. This ranking is based on multiple factors, including relevant work experience, educational qualifications, interview performance, and potential cultural fit. The HR Manager reviews the ranked list and makes the final hiring decision based on AI recommendations and human judgment. The system also generates detailed feedback reports for applicants, explaining their evaluation and areas for improvement. If two candidates receive similar scores, TrueSelect© highlights differentiating factors such as leadership potential or long-term commitment probability, helping HR teams finalize their decision. Unsuccessful candidates receive constructive feedback, improving their future job

prospects. If needed, candidates can request a secondary review to address concerns about bias or unfair evaluation, ensuring a transparent and just process.

AI act risk classification

In developing AI systems for recruitment, it is essential to address the legislative and compliance frameworks that govern their use, particularly as AI-based R&S systems are classified as high-risk under the EU AI Act. This classification arises from the significant societal impact of AI in employment-related domains, including CV-sorting, worker management, and access to self-employment opportunities¹¹(as stated on page 16, point 57 of the AI Act). High-risk classification brings with it a series of requirements aimed at ensuring fairness, transparency, and security in the R&S system's operation. Fulfilling these legal obligations is not only a matter of regulatory compliance but also a critical step building trust among stakeholders and end-users. Based on that European AI Act directive we classify R&S AI models as high-risk systems that would have to comply with multiple requirements to be deployed. **Those requirements include:**

- Implementation of risk management measures
- Obtain a data quality assessment
- Maintain proper code documentation and transparency
- Employ human oversight
- Ensure cybersecurity of the system
- Conduct conformity assessment with EU officials
- Uphold host-market monitoring and data governance
- Register the system in the EU database

Conclusion

The integration of AI into recruitment processes presents a promising opportunity for companies to enhance efficiency, reduce costs, and promote fairness in candidate selection. Throughout this domain analysis, we have examined the complexities of AI-driven recruitment, addressing key stakeholders, potential biases, legislative challenges, and system functionalities. The primary objective of *TrueSelect®* is to optimize hiring while maintaining transparency, compliance, and ethical standards.

From a **business perspective**, large tech companies benefit significantly from automating resume screening, interview assessments, and candidate ranking. The AI-driven system reduces HR workload, enhances decision-making, and improves candidate selection through data-driven insights. However, it is crucial that AI models remain explainable and adaptable to prevent unintended biases from influencing hiring outcomes.

From an **applicant's perspective**, AI recruitment tools must prioritize fairness, feedback, and accessibility. Providing applicants with detailed insights into their evaluations allows them to improve their skillsets and reapply with better qualifications. Ensuring transparency in AI decisions fosters trust and mitigates concerns about algorithmic discrimination.

From a **regulatory standpoint**, compliance with the **EU AI Act** and other relevant legislation is paramount. As recruitment AI systems are classified as high-risk applications, they must incorporate explainability, data governance, and continuous bias monitoring. Failure to meet these requirements could result in reputational damage, legal consequences, and decreased adoption rates.

¹¹<https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai>

Recommended Variables for AI Model Implementation

Based on our analysis, we recommend including the following **key variables** in the AI recruitment model:

- **Educational Variables:** Highest degree obtained, field of study relevance, certifications, and academic achievements.
- **Work Experience Variables:** Previous job titles, relevant experience, technical skills, and turnover rate.
- **Personal Attributes:** Interpersonal skills, problem-solving capabilities, and alignment with company culture.
- **Bias Mitigation Strategies:** AI bias audits, fairness tracking, and compliance with diversity requirements.

Final Considerations

While AI-driven recruitment has the potential to revolutionize hiring processes, it is not without its challenges. To ensure success, *TrueSelect©* should adopt a **hybrid approach**, where AI serves as an *augmenting tool* rather than a replacement for human decision-making. Continuous model refinement, human oversight, and transparency in decision-making will be essential in building a system that balances efficiency, fairness, and trust.

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